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Joint Migration Inversion of Vertical Seismic Profile Full-Wavefield Data: The First Application Offshore Abu Dhabi

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Abstract

Joint Migration Inversion is a joint and simultaneous implementation of velocity model building and imaging based on an iterative optimization process leveraging full-wavefields via a full-waveform matching scheme. The full-wavefields include not only primaries but also all multiples from both up- and down-going wavefields explained by the model parameters, namely velocity and reflectivity series. Therefore, the full-wavefields have a valuable opportunity to extend the subsurface illumination and refine both the velocity model and reflectivity image, even though dealing with data from a sparse and non-uniform acquisition geometry. In this paper, we applied this novel technology to 2D walk-away Vertical Seismic Profile data offshore Abu Dhabi with the objective to prove the concept and demonstrate the virtues. We confirmed that the technology can offer reliable solutions of both the velocity model and reflectivity image, and that the technology can considerably enhance the resulting data quality compared to conventional processing. In addition, we realized that the technology could considerably shorten the turnaround time thanks to no requirement of intensive pre-processing as well as its unified implementation of velocity model building and imaging.

Introduction

Vertical Seismic Profile (VSP) surveys, as well as surface seismic ones, play a critical role in the industry to image subsurface structures, characterize subsurface properties, thus understand reservoirs and their behaviors. VSP surveys provide broader frequency-band data than surface seismic ones, which results in higher-resolution images around boreholes. This is mainly because of positioning of receivers in a borehole while positioning of sources at the surface, which allows shot-generated wavefields to travel the subsurface only once. However, conventional VSP processing commonly utilizes only primary reflections, after many pre-processing steps such as denoise, deconvolution and demultiple, which consequently generates the reflectivity image just around the borehole. Moreover, the velocity model is usually built from check-shot data again just around the borehole. In addition, it often requires a long turnaround time due to the complicated pre-processing steps.

Berkhout (2012 and 2014a) argued that seismic data are nonlinear in subsurface properties by definition and can be parameterized by nonlinearity properties such as velocity and reflectivity. He also claimed that full-wavefields include not only primaries but also all multiples from both up- and down-going wavefields, again explained by the nonlinearity properties, thereby have a valuable opportunity to extend the subsurface illumination and refine both the velocity model and reflectivity image. This is of particular importance when dealing with data from a sparse and non-uniform acquisition geometry, like VSP data. Full-wavefield approaches utilizing multiples in non-surface seismic data have been discussed for a few decades. For example, Reiter et al. (1991) introduced the concept for ocean-bottom-node data, and He et al. (2007) extended this to VSP data. However, ones utilizing primaries as well, handling cross-talks among primaries and multiples, updating a velocity model simultaneously, based on a large-scale inversion process, have not yet been established.

Berkhout (2014c) introduced Joint Migration Inversion (JMI), which is a joint and simultaneous implementation of velocity model building and imaging based on an iterative optimization process leveraging full-wavefields via a full-waveform matching scheme. Here, the full-wavefields are explained by the propagation and scattering effects, the former to describe the kinematics (i.e. phase or velocity) and the latter the dynamics (i.e. amplitude or reflectivity). In fact, JMI is an extension from Full-Wavefield Migration (FWM, Berkhout, 2014b). FWM is an imaging process reformulated first in a full-wavefield manner and second in a least-squares-inversion scheme. Berkhout (2014a and 2014b) introduced the background theory of FWM. Davydenko and Verschuur discussed the practical aspects (2017) and demonstrated some field data examples (2018). Soni and Verschuur (2014) applied FWM to 2D VSP data. JMI then includes a velocity-model-building process based on a reflection-tomography scheme on-the-fly as part of the imaging process. Berkhout (2014c) and Verschuur et al. (2016) introduced the background theory of JMI. Staal et al. (2014) proved the concept on 2D surface seismic data, while El-Marhfoul and Verschuur (2014, 2016, 2017 and 2019) and Al-Bannagi et al. (2018) extended this to 3D VSP data. In this paper, we apply JMI to 2D walk-away VSP data in order to demonstrate the virtues compared to conventional VSP processing.

Method

JMI is an iterative optimization process using a closed loop to update the model parameters (i.e. velocity and reflectivity series) jointly and simultaneously (Fig. 1). It leverages full-wavefields in the full waveform in such a way that the observed data are fully explained by the model parameters. Here, all surface-related and internal multiples can be included in the process. Their cross-talks are naturally removed from the reflectivity image, while their energy is accordingly focused into right positions in the image domain. In this way, multiples can contribute to extending the subsurface illumination that primaries cannot cover. Note that no demultiple is thus required for the observed data in the pre-processing phase.

In the closed loop, the forward-modelled data are reconstructed from the model parameters in a recursive manner and compared with the observed data. The residue is translated into the velocity and reflectivity gradients to update the model parameters. By closing the loop and feeding back the residue in the engine, the model parameters are iteratively updated and optimized based on a gradient-descent scheme. For the reflectivity updates, the objective function in Fig. 1 is minimized until it reaches the global minimum. Here, P_{obs} is the observed data at the measurement surface z_0, and P_{mod} the forward-modelled data. The subscript ω denotes the summation over all frequencies, and the subscript k the summation over all shot gathers. On the other hand, for the velocity updates, the cross-correlation between the forward-modelled data and the observed data is maximized as it aims for the global maximum. The combination of different objective functions is indeed helpful to converge to more accurate solutions. Consequently, the outputs are the velocity model and reflectivity image that fully explain the observed data in terms of both the kinematics and dynamics.

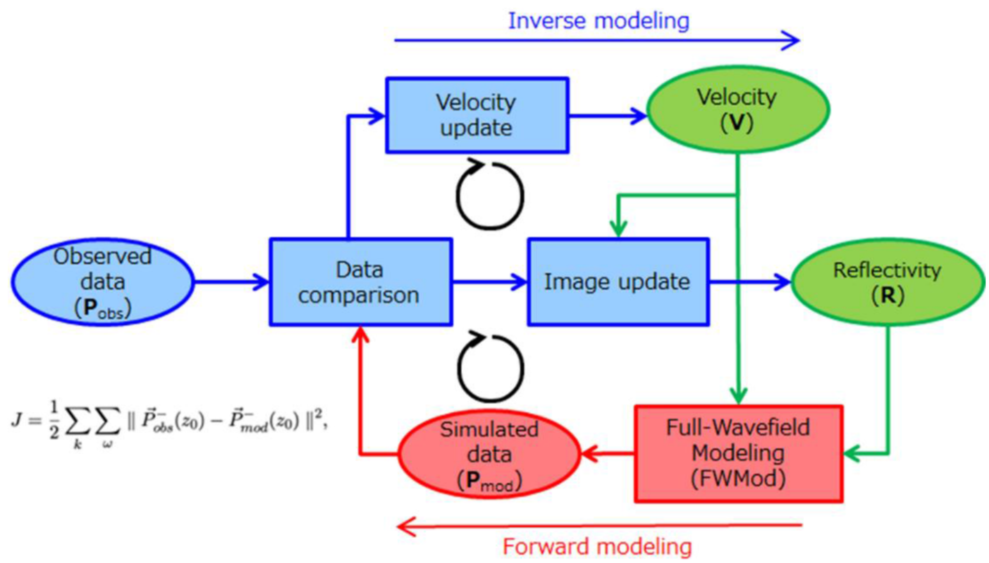

Figure 1—Closed loop for the iterative optimization process of JMI. The model parameters are in green, the forward modelling modules in red, and the inverse modelling modules in blue. *J* is the objective function.

JMI is valid for any acquisition geometry including a reciprocal case of VSP, in which reciprocal sources are positioned in the borehole and reciprocal receivers are located at the surface. One of the advantages of VSP data is the availability of the observed direct wavefield representing the Green's function for every common receiver gather (i.e. at every reciprocal source position). The direct wavefield can be focused at the corresponding depth position so that the reciprocal source wavelet can be estimated there based on a certain velocity model. Therefore, source wavelets can also be included as the model parameters in the process, thus iteratively updated and optimized based on an updated velocity model. Better source wavelets result in a more accurate velocity model and vice versa.

Field data

We applied JMI to 2D walk-away VSP data sets offshore Abu Dhabi. The VSP survey was originally acquired for four azimuth lines (in addition to one walk-around circle), each of which consists of several source lines for different tool levels in the borehole (Fig. 2). The line of azimuth 117 ºN consists of five source lines for five tool levels to cover depths from the reservoir to the overburden. The other lines at azimuths 28 ºN, 72 ºN and 163 ºN consist of two source lines for two deepest tool levels to cover depths only around the reservoir. The tool levels are not continuous, thereby there are some gaps between them, which results in a sparse and non-uniform acquisition geometry. The shot point interval at the surface is 25 m in nominal. The tool is configured with 20 geophones with a spacing of 15 m. The borehole is deviated with the maximum deviation of about 20 °.

The 2D VSP data sets were originally processed using conventional steps such as denoise, data orientation, wavefield separation, deconvolution, demultiple, velocity model building, followed by migration imaging. In addition, the vertical and horizontal transverse anisotropic parameters (VTI and HTI) and quality factor (Q) were estimated.

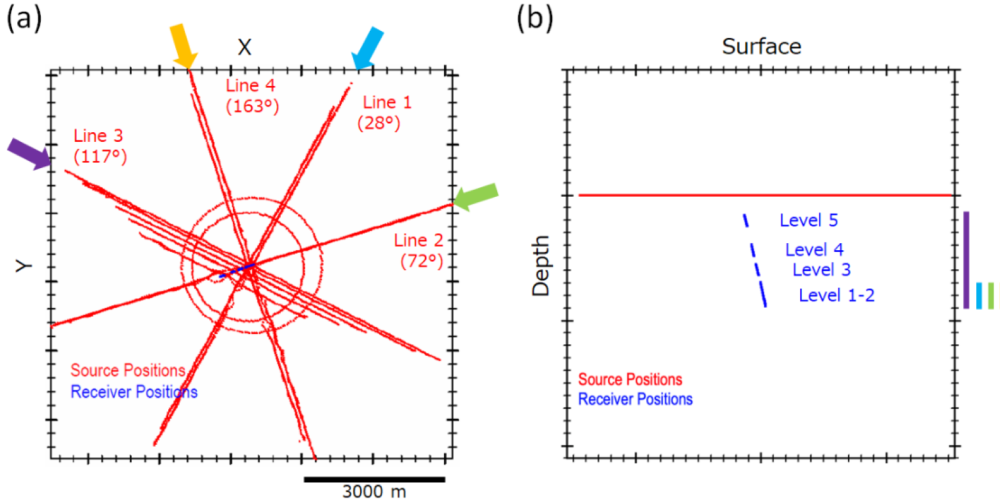

Figure 2—Acquisition geometry of the VSP survey: (a) map view; (b) section view along the line of azimuth 117 °N. The source positions are in red, and the receiver positions in blue. The source lines and receiver coverages of azimuths 117 °N, 28 °N, 72 °N and 163 °N are in violet, cyan, green and orange, respectively. The two deepest tool levels are positioned around the reservoir.

For this work, the 2D VSP data sets were processed using JMI, each azimuth line of which was independently processed in a 2D manner. The inputs were the observed data of up- and down-going wavefields after pre-conditioning steps of up-/down-going-wavefield decomposition followed by simple denoise and regularization in the common receiver domain. The up-going wavefield mainly contains primary reflections, while the down-going wavefield holds direct waves, as well as multiples. For the regularization, a matching pursuit scheme with Anti-Leakage Fourier Transform (ALFT) was applied. See, for example, Xu et al. (2005 and 2010) and Jahanjooy (2014). Regarding the iterative optimization process to update the model parameters, first, only the reflectivity series were updated with an initial 1.5D velocity model from the well log and check-shot data. After several iterations, both the velocity and reflectivity series, in addition to the source wavelets in the reciprocal domain, were updated and optimized in a flip-flop manner. For the forward modelling, the VTI and Q effects were optionally included in the wavefield propagation and scattering. The outputs were the velocity model and reflectivity image to explain the full-wavefields in the observed data, in addition to the ones to explain only primaries solely for comparison.

Results and comparison with legacy ones

Fig. 3, Fig. 4, Fig. 5 and Fig. 6 show the conventional VSP processing and JMI results along the line of azimuth 117 °N. Table 1 summarizes the quantitative attributes such as amplitude spectra and signal-to-noise ratios (S/Ns) for these results. Fig. 3 presents the reflectivity images. First, the conventional migration image (Fig. 3a) is vertically and horizontally limited just around the borehole. Moreover, it contains dominant migration smiles due to the poor focusing. Next, the JMI image from only primaries (Fig. 3b) is quite similar to the first one. In contrast, the JMI image from full-wavefields (Fig. 3c) is significantly spread away from the borehole due to the extended subsurface illumination. Furthermore, it is concordant with the subsurface structure defined by the horizons in the region. In addition, it does not contain a kind of migration smiles, thus reveals a high S/N. Hereinafter in this paper, "JMI image" refers to the one from full-wavefields unless otherwise mentioned. Last, the JMI image with Q compensation (Fig. 3d) contain a broader frequency bandwidth, thus further enhances the resolution.

Table 1—Resulting quantitative attributes for the conventional migration and JMI images.

Processing	Conventional	JMI with full-wavefields			
Azimuth	117 °N	117 °N	28 °N	72 °N	163 °N
S/N (dB)	8.12	27.30	25.83	26.89	25.87
Well-to-seismic correlation	0.69	0.85	0.76	0.57	0.39

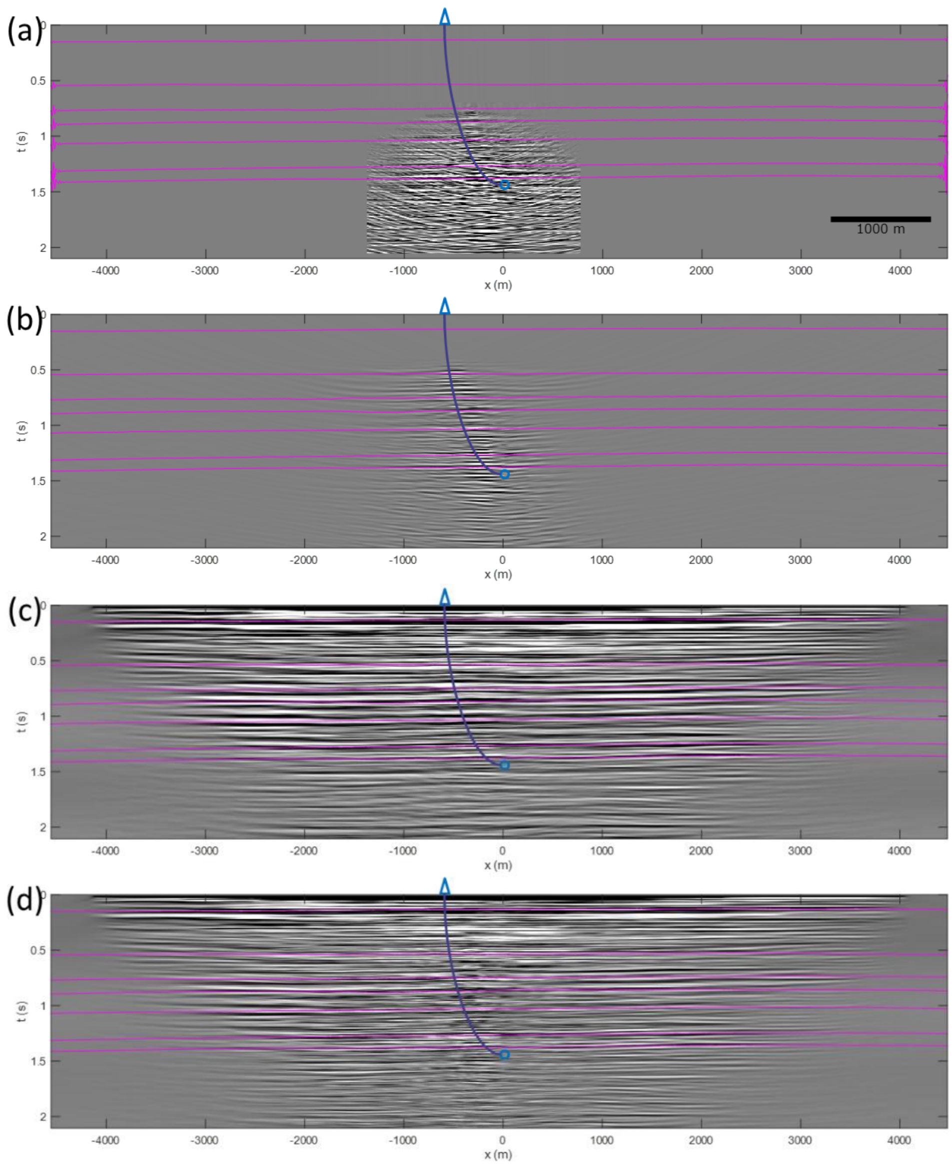

Figure 3—Conventional migration and JMI images along the line of azimuth 117 °N: (a) conventional migration image; (b) JMI image from only primaries; (c) JMI image from full-wavefields; (d) JMI image from full-wavefields with Q compensation. The well trajectory is projected on the section by a navy-blue curve. Key horizons in the region are superimposed on the section by pink lines. The two deepest horizons are around the reservoir.

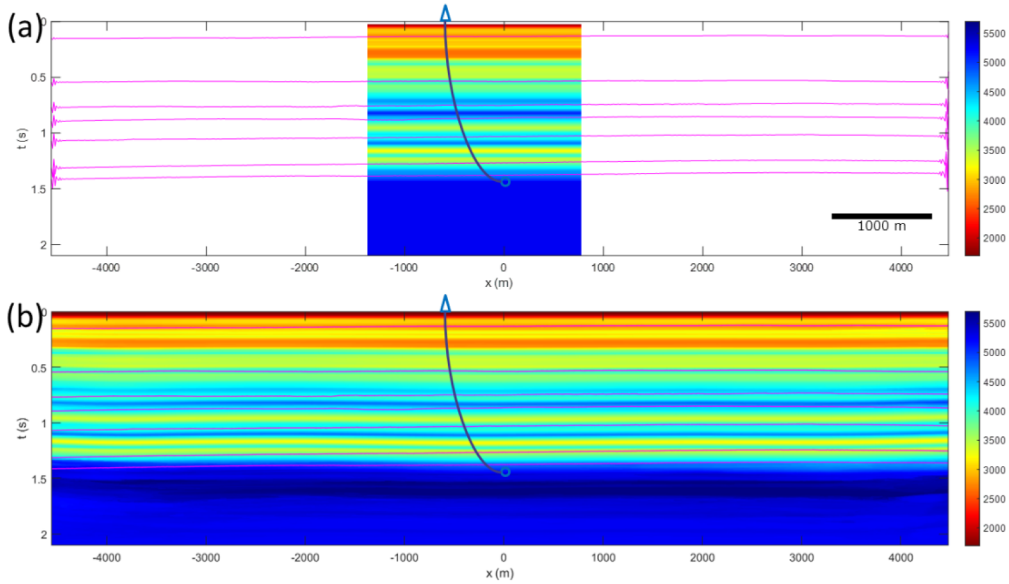

Figure 4—(a) Conventional velocity model; (b) JMI velocity model along the line of azimuth 117 °N. The well trajectory is projected on the section by a navy-blue curve. Key horizons in the region are superimposed on the section by pink lines. The two deepest horizons are around the reservoir.

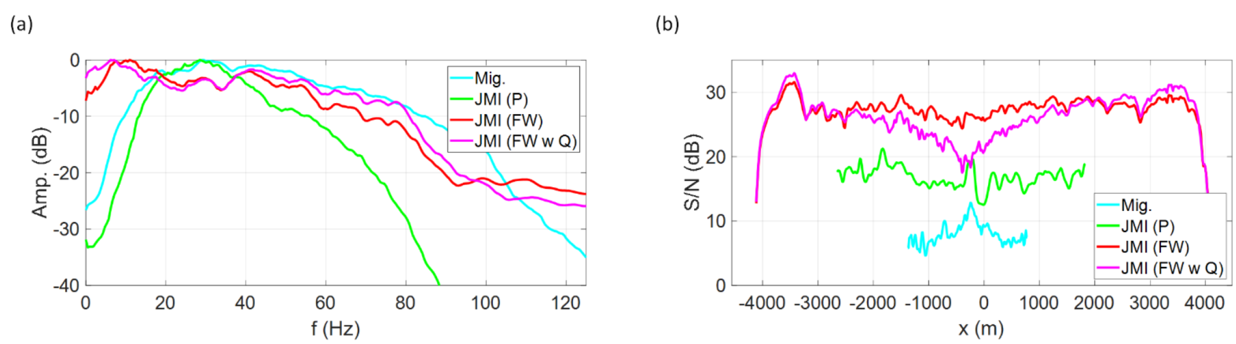

Figure 5—(a) Amplitude spectra; (b) S/Ns for conventional migration and JMI images along the line of azimuth 117 °N.

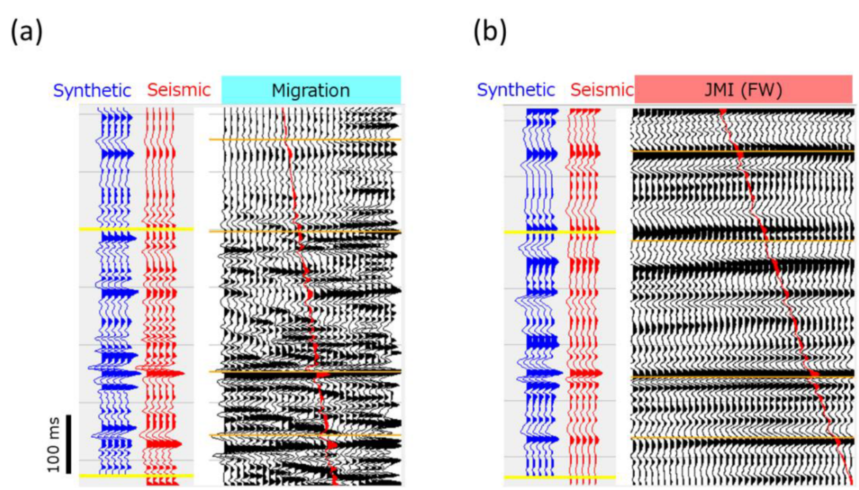

Figure 6—Well-to-seismic tie along the line of azimuth 117 °N: (a) conventional migration image; (b) JMI image. The synthetic seismogram is in blue, and the seismic trace along the well trajectory in red. The correlation window is denoted by yellow lines. Key well markers are superimposed on the section by orange lines. The two deepest well markers are around the reservoir.

Fig. 4 presents the corresponding velocity models. The conventional velocity model (Fig. 4a) is a simple 1.5D model built solely from the well log and check-shot data, thus limited again just around the borehole. Besides, the JMI velocity model (Fig. 4b) is significantly spread in concordant with the subsurface structure as well as the JMI image.

Fig. 5 presents the amplitude spectra and S/Ns, respectively, for the conventional migration and JMI images. The JMI image has a broader frequency bandwidth, especially for low frequencies less than 20 Hz and high frequencies more than 100 Hz, compared to the conventional migration image (Fig. 5a). Moreover, the JMI image has a much higher S/N up to the average value of about 27 dB compared to the conventional migration image with the average value of about 8 dB (Fig. 5b).

Fig. 6 presents the well-to-seismic ties for the conventional migration and JMI images. The conventional migration image has a fair well-tie with the correlation value of about 0.7 probably due to the residual multiples and dominant migration smiles, while the JMI image has a better well-tie up to the correlation values of about 0.9. All the above outcomes indicate that JMI considerably enhances the resulting data quality compared to conventional VSP processing.

Fig. 7, Fig. 8, Fig. 9 and Fig. 10 show the JMI results along all the azimuth lines. See also Table 1. Remember that the line of azimuth 117 °N is composed of five tool levels to cover depths from the reservoir to the overburden, while the other azimuth lines are composed of two deepest tool levels to cover depths only around the reservoir. Therefore, the line of azimuth 117 °N comprises a wider receiver coverage from deep to shallow levels than the other azimuth lines. Fig. 7 presents the reflectivity images extracted in the vicinity of the borehole. These reflectivity images are quite similar with each other, thus comparable at the intersections at all depths. Fig. 8 presents the corresponding velocity models. Again, these velocity models closely resemble with each other, thus consistent at the intersections at any depth. Fig. 9 presents the amplitude spectra and S/Ns, respectively, for all the azimuth lines. The amplitude spectra show a quite similar frequency bandwidth and resolution for all the azimuths. The same holds true for S/Ns regardless of the azimuth. These indicate that JMI advantageously recovers reliable solutions of velocity model and reflectivity image regardless of the receiver coverage because of not only horizontally but also vertically extending the subsurface illumination.

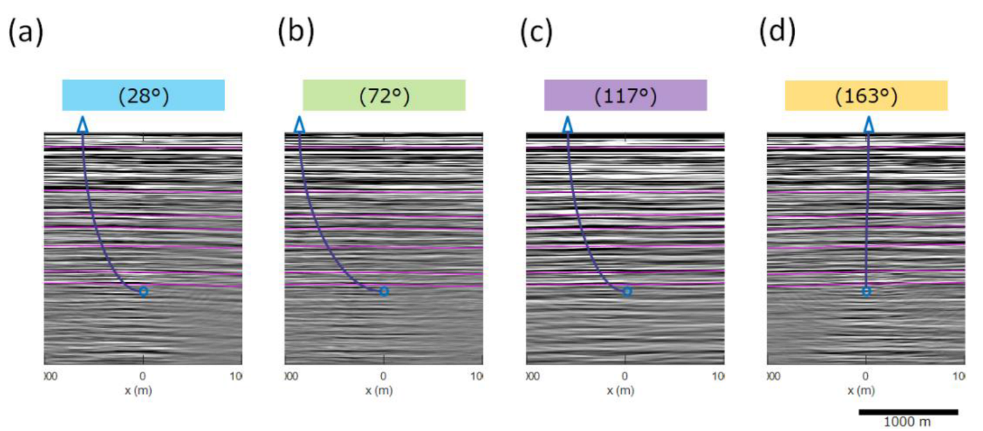

Figure 7—JMI images along the four azimuth lines: (a) azimuth 28 °N; (b) azimuth 72 °N; (c) azimuth 117 °N; (d) azimuth 163 °N. The well trajectory is projected on the section by a navy-blue curve. Key horizons in the region are superimposed on the section by pink lines. The two deepest horizons are around the reservoir. The coordinate origin in x (m) is at the intersection of all the azimuth lines.

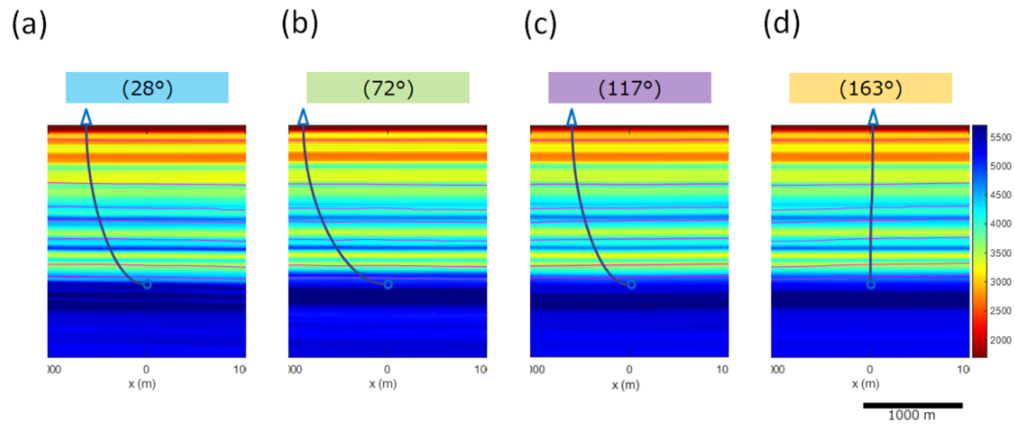

Figure 8—JMI velocity models along the four azimuth lines: (a) azimuth 28 °N; (b) azimuth 72 °N; (c) azimuth 117 °N; (d) azimuth 163 °N. The well trajectory is projected on the section by a navy-blue curve. Key horizons in the region are superimposed on the section by pink lines. The two deepest horizons are around the reservoir. The coordinate origin in x (m) is at the intersection of all the azimuth lines.

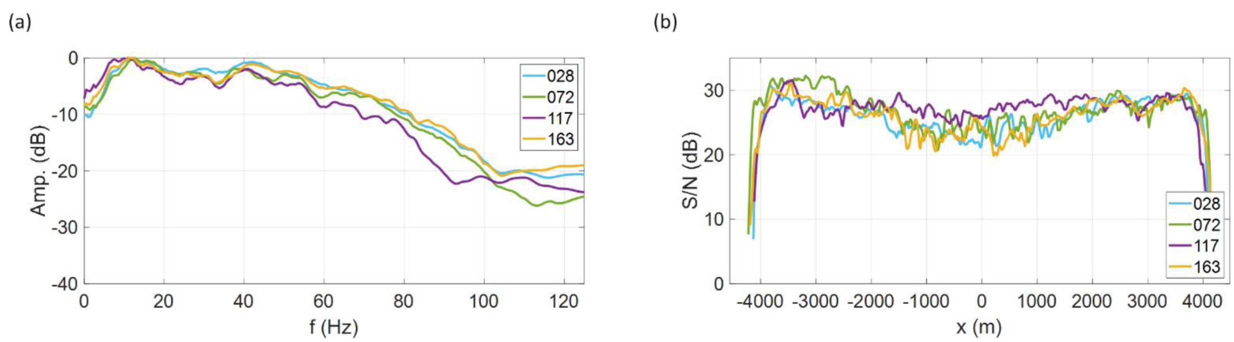

Figure 9—(a) Amplitude spectra; (b) S/Ns for JMI images along the four azimuth lines.

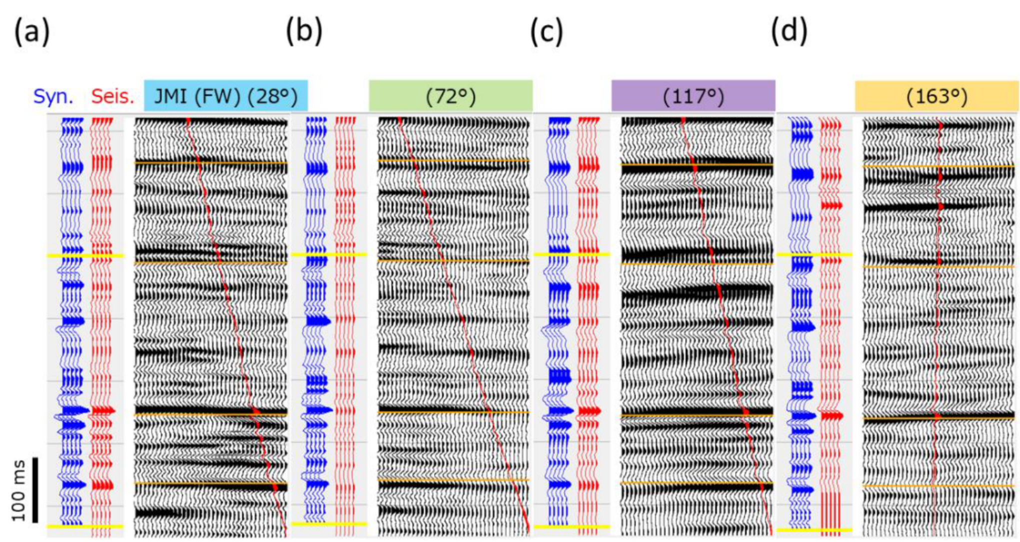

Figure 10—Well-to-seismic tie along the four azimuth lines: (a) azimuth 28 °N; (b) azimuth 72 °N; (c) azimuth 117 °N; (d) azimuth 163 °N. The synthetic seismogram is in blue, and the seismic trace along the well trajectory in red. The correlation window is denoted by yellow lines. Key well markers are superimposed on the section by orange lines. The two deepest well markers are around the reservoir.

Fig. 10 presents the well-to-seismic ties for all the azimuth lines. The line of azimuth 117 °N shows the best well-tie up to the correlation value of about 0.9, while the other azimuth lines show a fair well-tie with lower correlation values probably due to the limited receiver coverage. This indicates that JMI could still converge to more accurate solutions, thus further improve the resulting data quality, if it can utilize more data available in the process.

Conclusions

We applied JMI to 2D walk-away VSP data offshore Abu Dhabi. This led to the following conclusions:

1. JMI offers reliable solutions of both the velocity model and reflectivity image even though dealing with data from a sparse and non-uniform VSP acquisition geometry.
2. JMI considerably enhances the resulting data quality compared to conventional VSP processing: (i) extended subsurface illumination, thus wider velocity model and image area in more concordant with the subsurface structure; (ii) higher resolution and S/N; (iii) better well-tie. This is mainly because of leveraging full-wavefields in the full waveform, in which multiples are utilized for the subsurface illumination to reinforce mapping of primaries even away from the borehole.
3. JMI could still converge to more accurate solutions, thus further improve the resulting data quality, if it can utilize more data available in the process. For instance, JMI for 3D Distributed Acoustic Sensing (DAS) VSP can do so since 3D VSP can extend the source coverage and DAS the receiver coverage in the acquisition.
4. JMI could considerably shorten the turnaround time thanks to no requirement of intensive pre-processing as well as its unified implementation of velocity model building and imaging.

Since we successfully proved the concept, demonstrated the virtues and unlock the great potential, we are committed to using this novel technology in our future VSP operations.

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